IDC 203: INTRODUCTION TO EARTH SCIENCES

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Outline

1. Glacial: the work of ice

2. Winds as weathering agents

Glaciers: the Work of Ice

Glacier

- Body of ice, snow, firn, and meltwater lying wholly or mostly on land showing evidence of present or former motion.

- Glaciers form where more snow accumulates in winter than melts in summer.

Types of Glaciers

- Based on their formation
- Geophysical classification
- Morphological classification

Types of Glaciers

- Based on their formation
  - Alpine: restricted to mountainous systems
    - Himalaya, Alps
  - Continental: covers extremely large areas
    - Antarctica: 12,500,000 km2 up to 3 km thick, 80% of the ice on earth, 65% of the fresh water
    - Greenland: 1,700,000 km2 3.2 km thick

Types of Glaciers

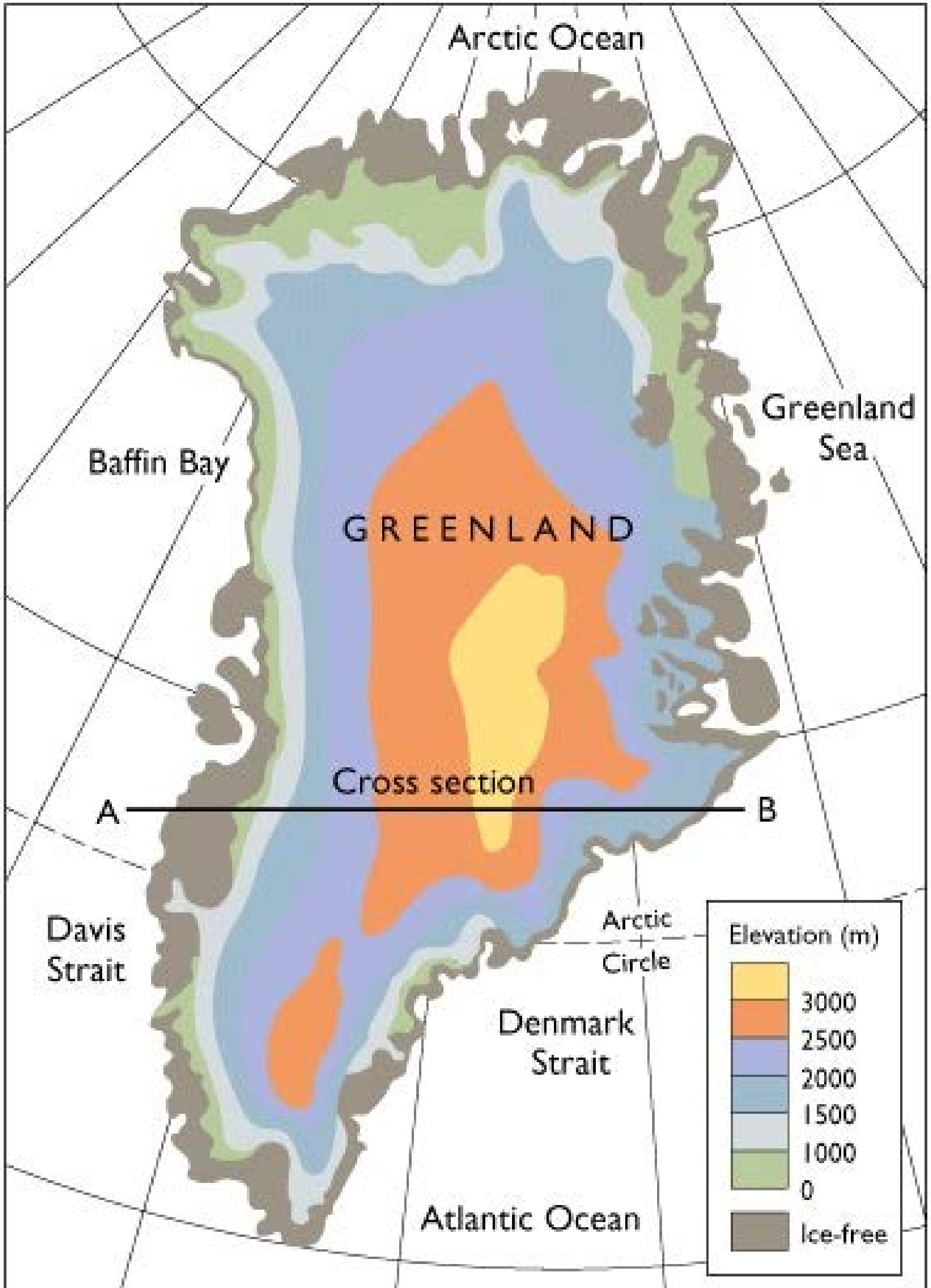
- Geophysical classification
  - Warm base - water at the bottom
    - Alps
  - Cold base - ice at the bottom
    - Antarctic and Greenland ice sheet

Types of Glaciers

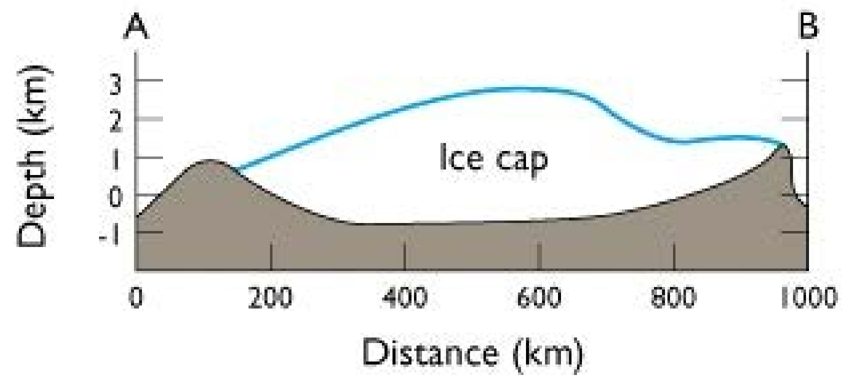
- Morphological classification
  - Valley glacier
    - Valley glaciers are streams of flowing ice that are confined within steep walled valley
  - Ice cap glacier
  - Ice shelf glacier

Types of Glaciers

- Morphological classification
  - Valley glacier
  - Ice cap glacier
    - Ice caps are miniature ice sheets, covering less than 50,000 square kilometers (19,305 square miles) thin steep walled valley
  - Ice shelf glacier



Greenland Ice Cap



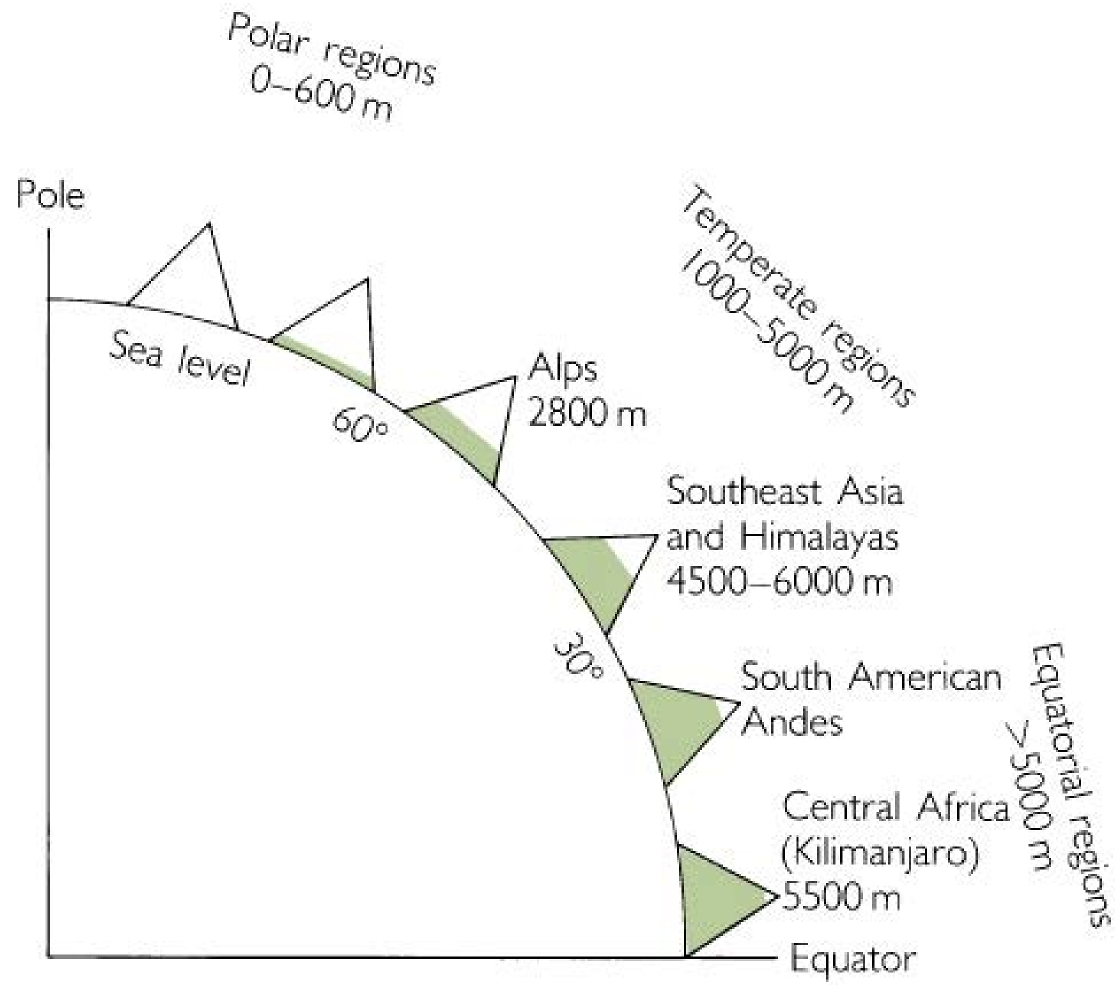
Types of Glaciers

- Morphological classification
  - Valley glacier
  - Ice cap glacier

  - Ice shelf glacier
    - An ice sheet is a chunk of glacier ice that covers the land surrounding it and is greater than 50,000 kilometers (20,000 miles) wide.

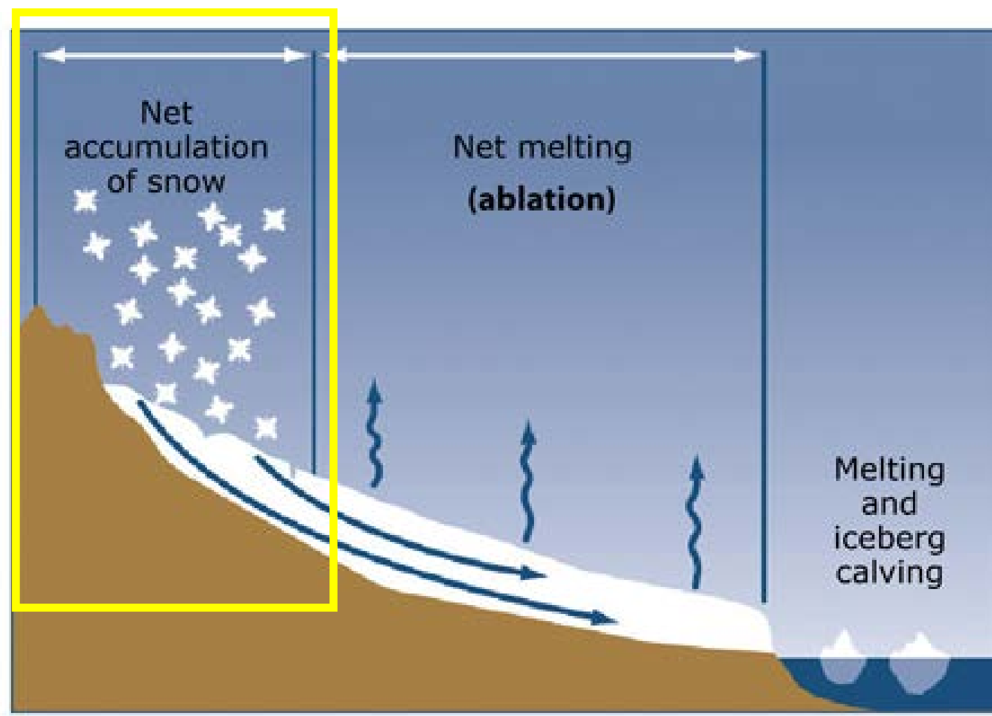

Sentinel Range, Antarctica

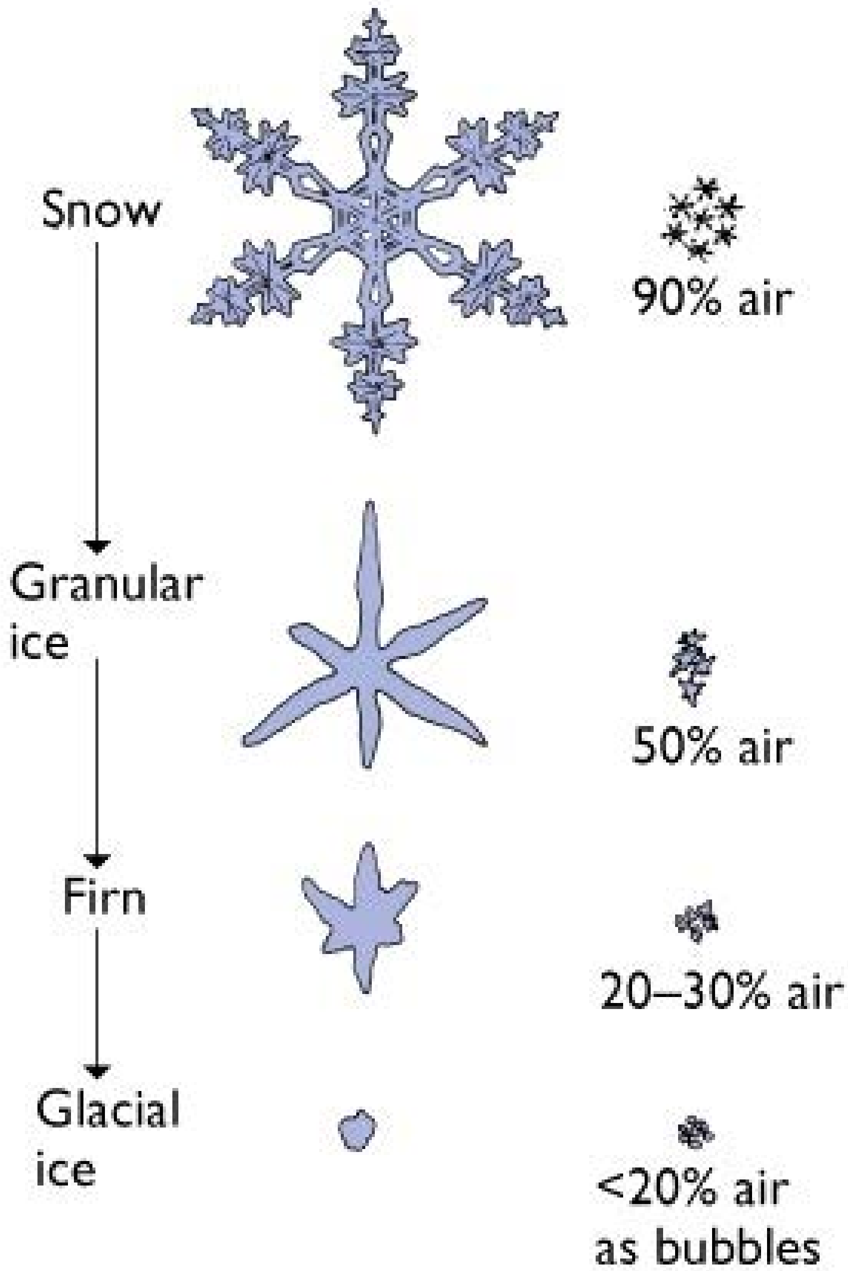
Variations in the Present Height of the Snow Line



Glacial growth: Accumulation

- Transformation of snow into ice
  - *Snow* may be thought of as sediment,
  - *Firn* as sedimentary rock, and
  - *Glacier ice* as metamorphic rock.

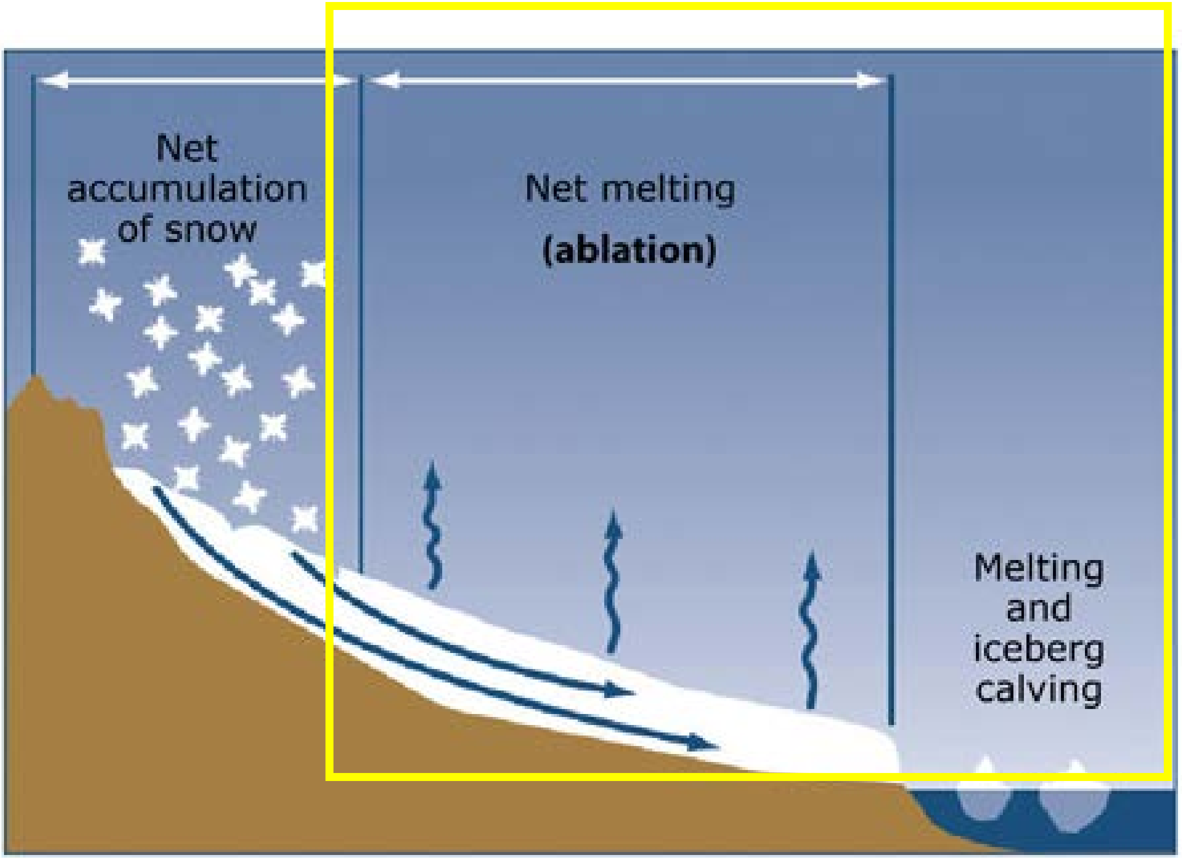




Transformation of Snow to Glacial Ice

Glacial shrinkage: Ablation

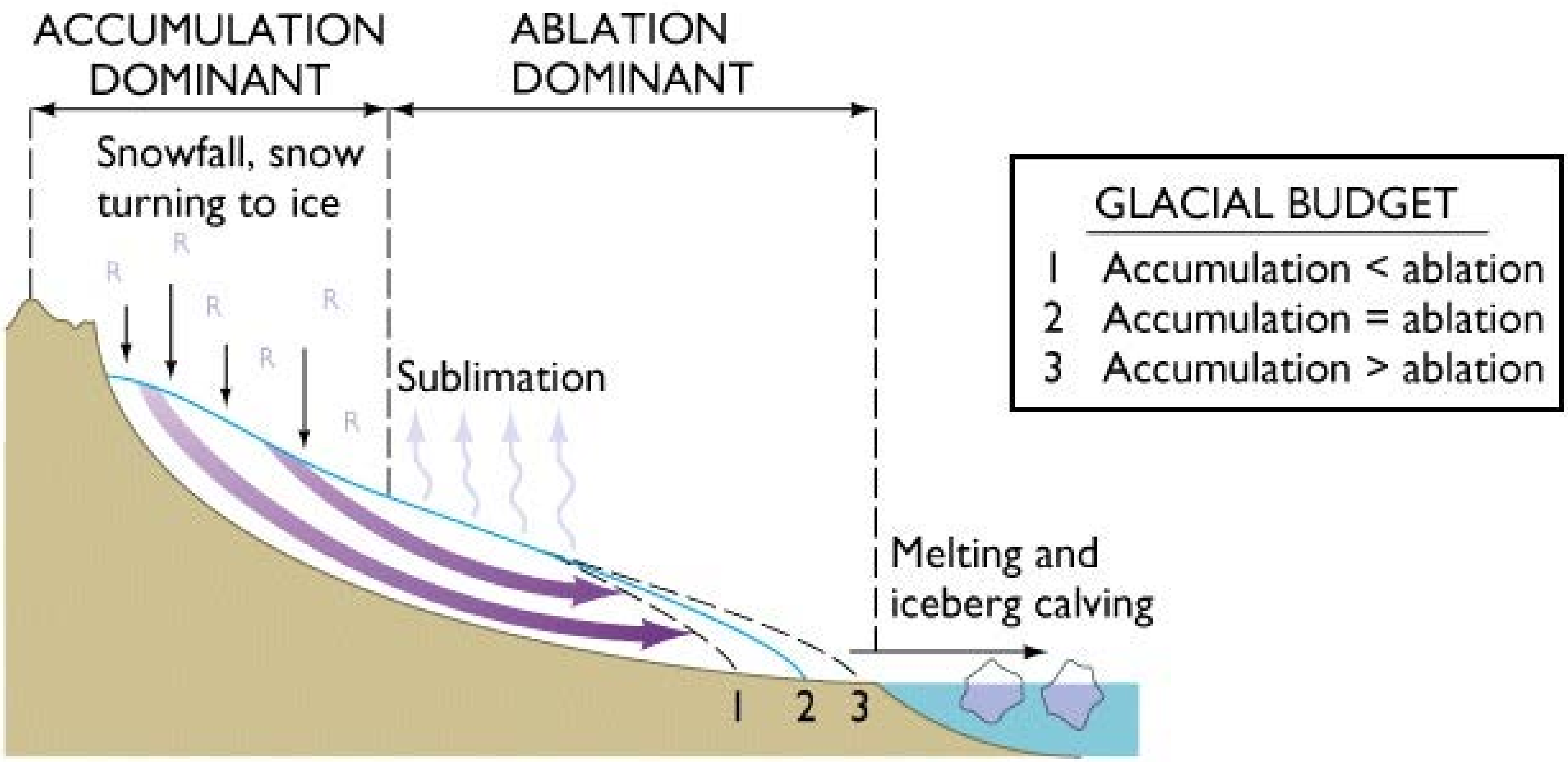
- Melting
- Iceberg calving: breaking off of chunks of ice at the edge of a glacier
- Sublimation



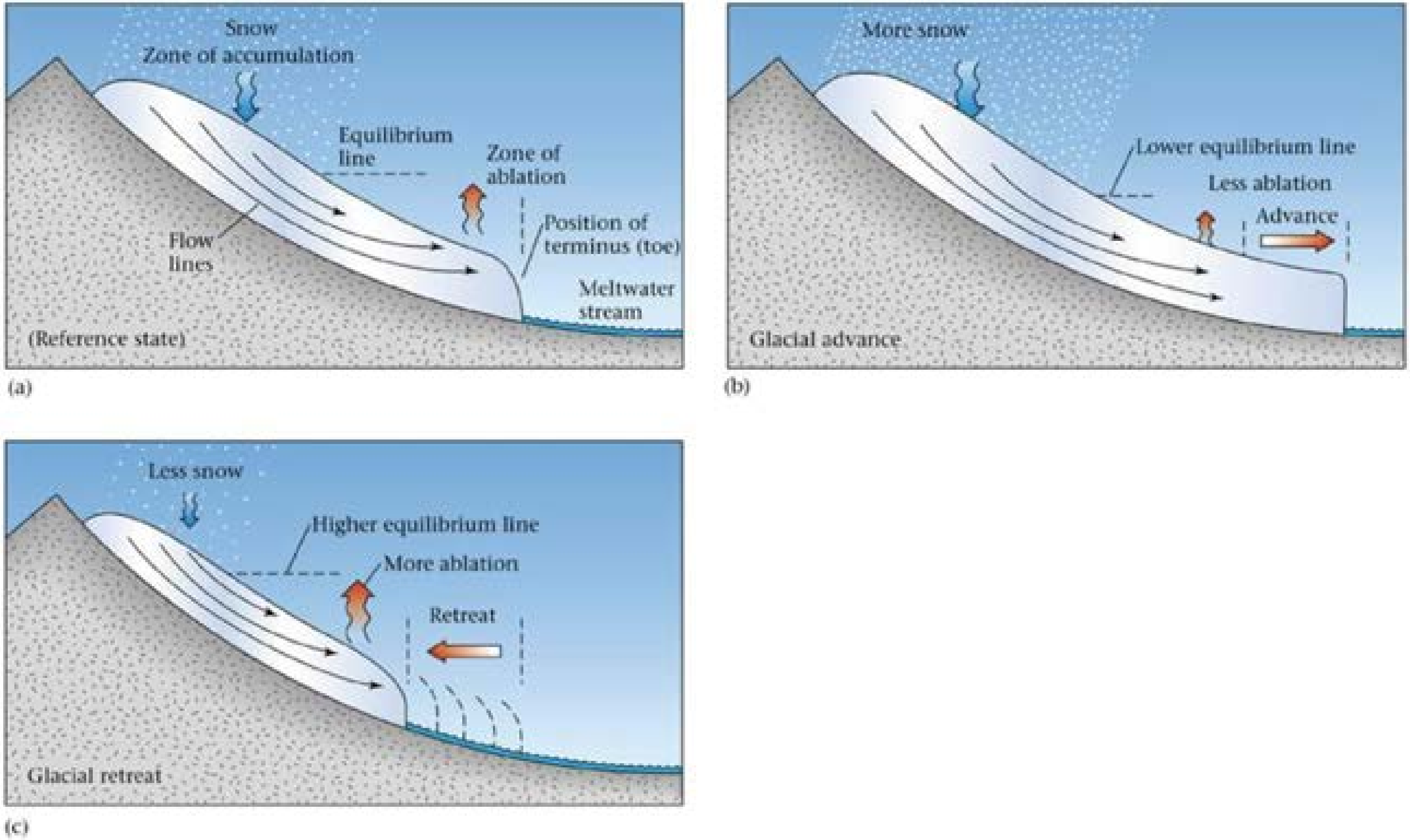
Calving Glacier, Glacier Bay, Alaska

Tom

Glacial Budget



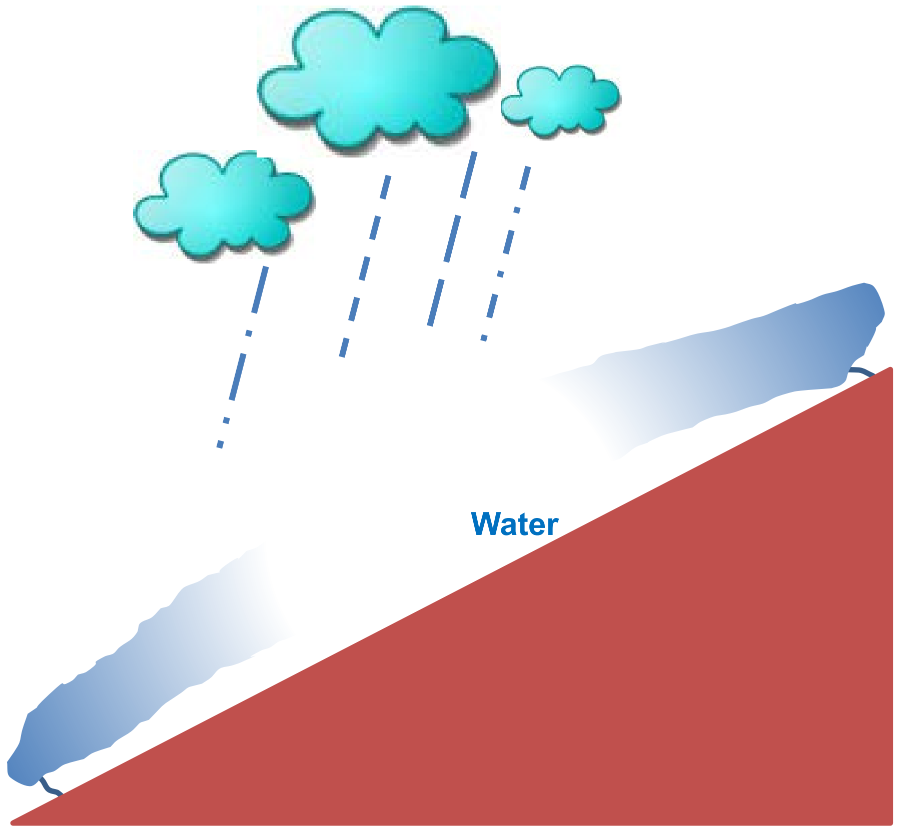
Glacial advance/retreat



Earth: Portrait of a Planet, 2nd Edition

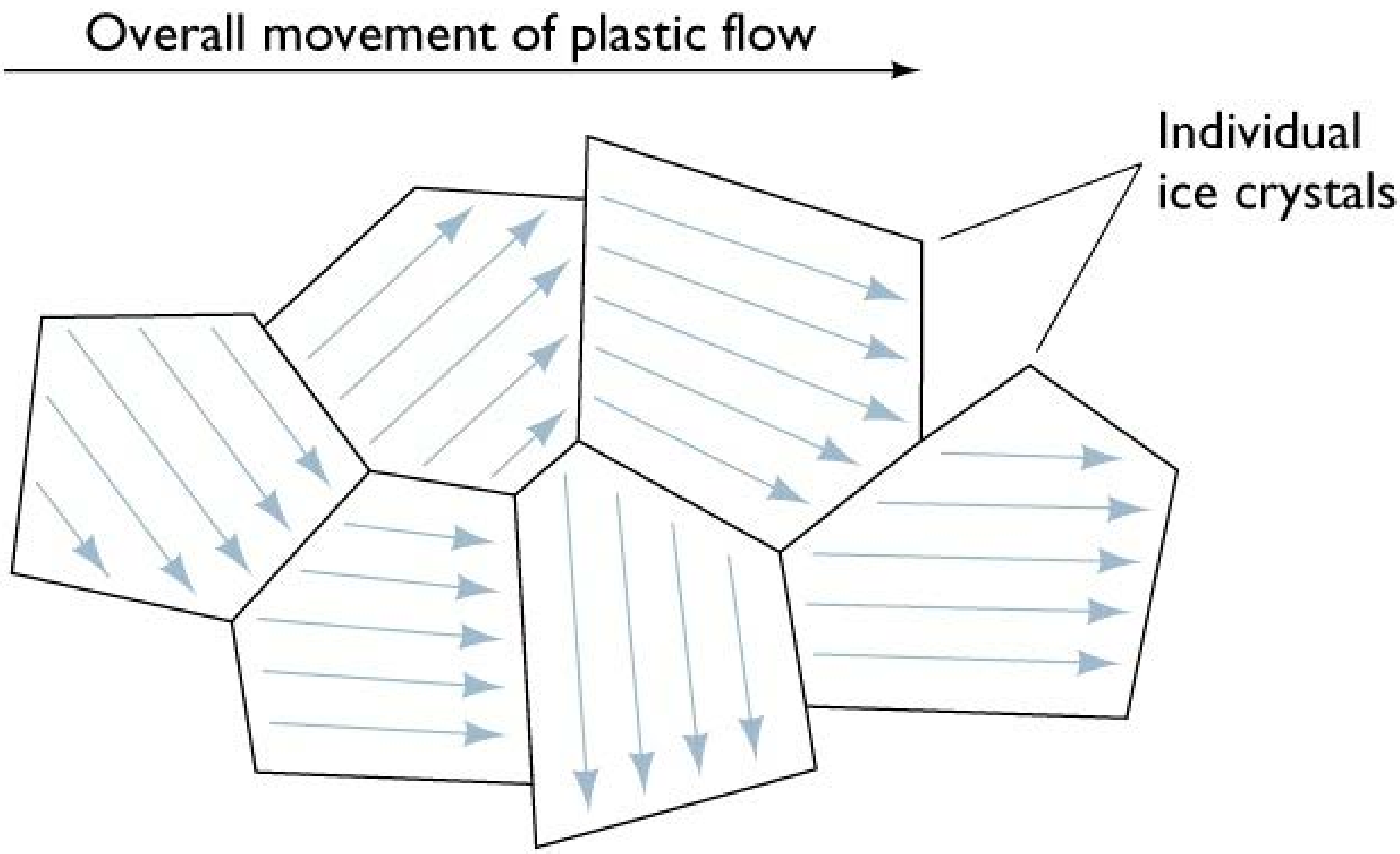
Mechanisms of motion in glaciers

- Rates of glacial movement depends
  - Slope steepness
  - Precipitation
  - Air temperature



Mechanisms of motion in glaciers

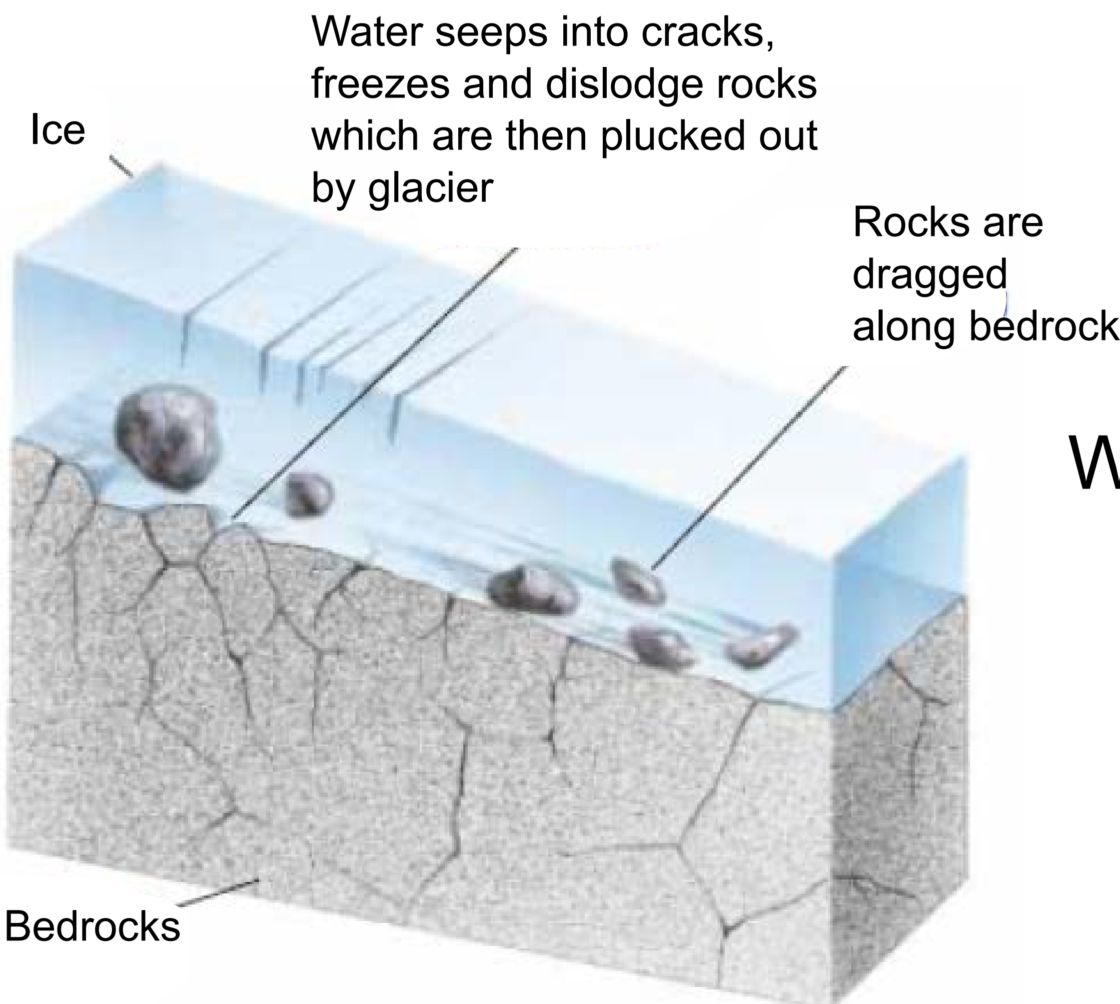
- Types of motion
  - Basal slip
    - Water accumulation
  - Plastic flow
    - Viscous flow


Overall movement of plastic flow
Individual ice crystals

Rates of motion

- Extremely variable from one glacier to another
- Millimeters to meters per day
- Some glaciers move in surges: periods of rapid movement following periods of quiescence.
- During surge, rates may be 50 m/day.
- Rates vary with position in the glacier.

Glacial erosion



Weathering processes

- Temperature change
- pressure release fracture

Glacial erosion

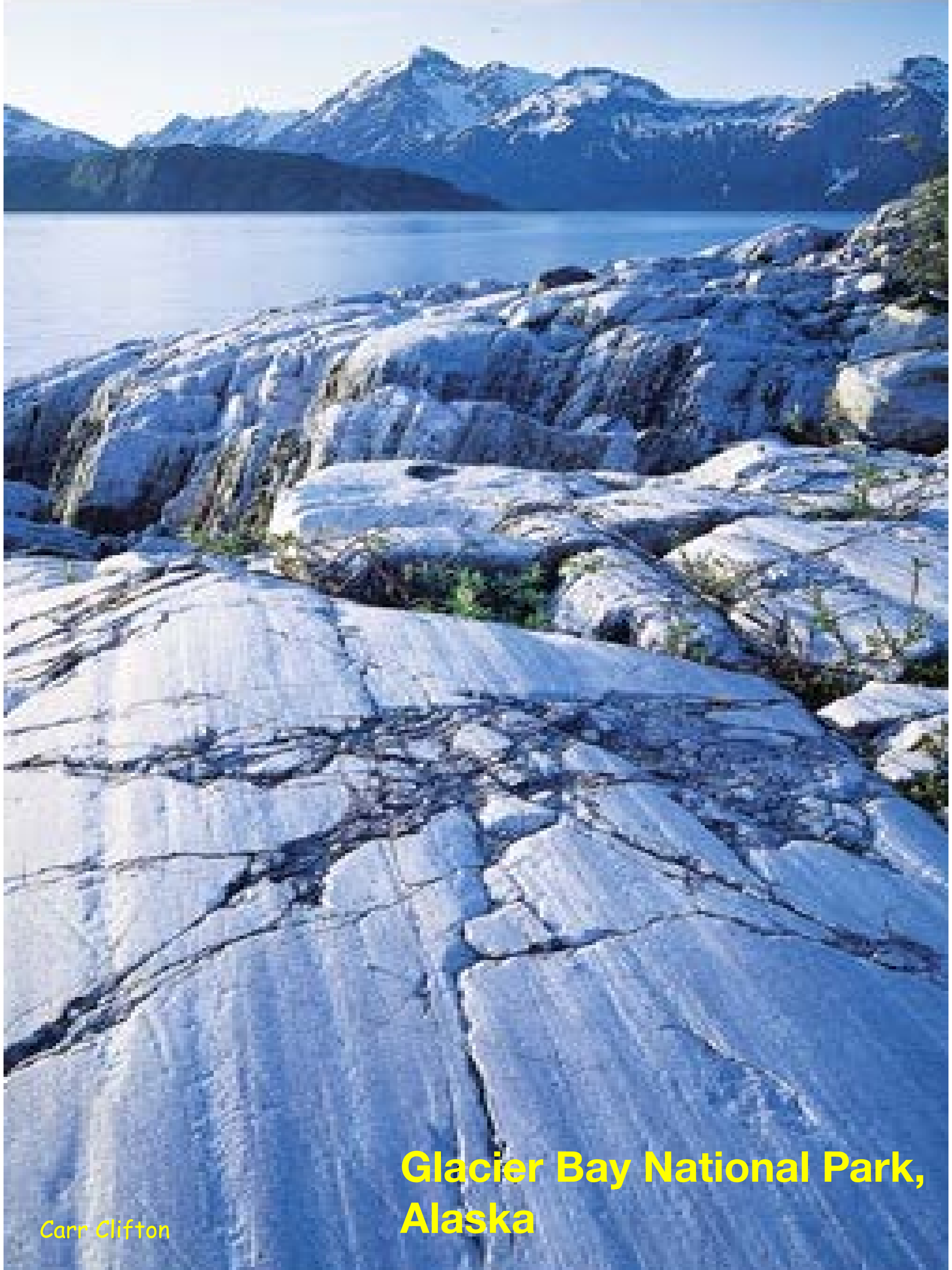
- Glacial erosional features
  - **Plucking**
    - rocks and stones become frozen to the base or sides of the glacier and are plucked from the ground or rock face as the glacier moves

Glacial erosion

- Glacial erosional features
  - **Glacial striations**
    - scratches or gouges cut into bedrock by glacial abrasion

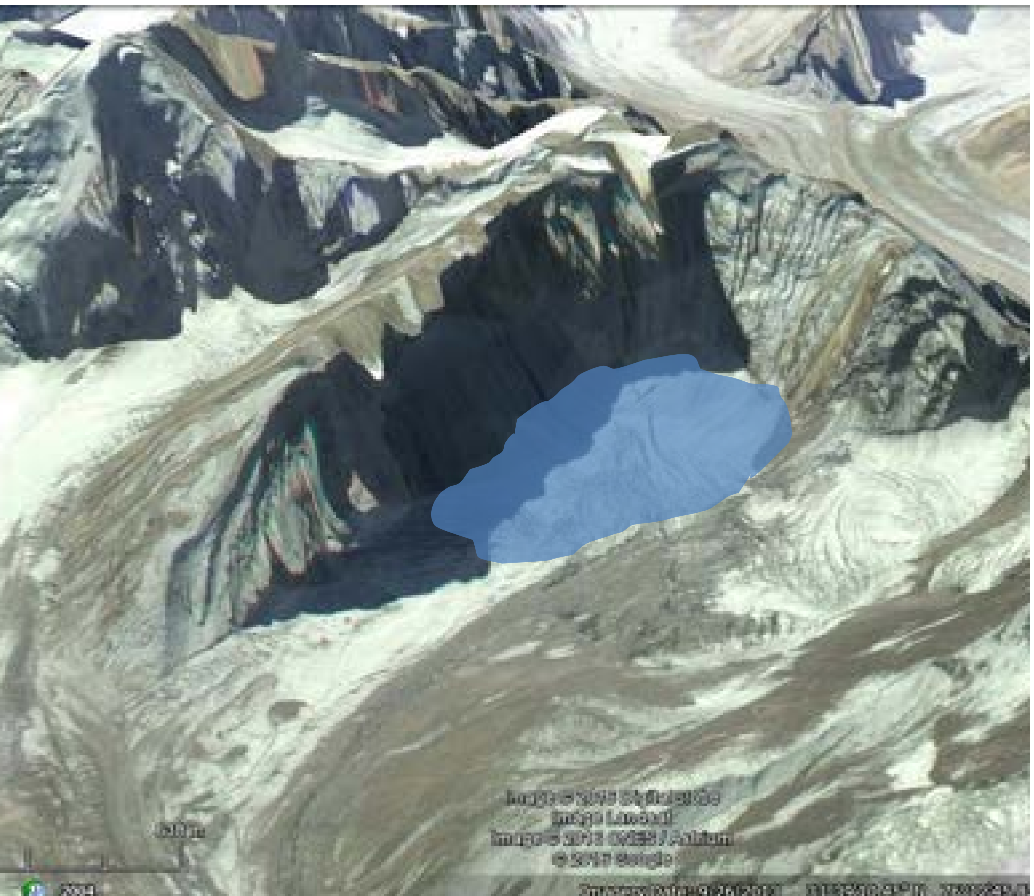
Glacial erosion

- **Rock flour**
  - fine-grained, silt-sized particles of rock, generated by mechanical grinding of bedrock by glacial erosion


Carr Clifton
Glacier Bay National Park, Alaska

Glacial Landforms

1. Cirque



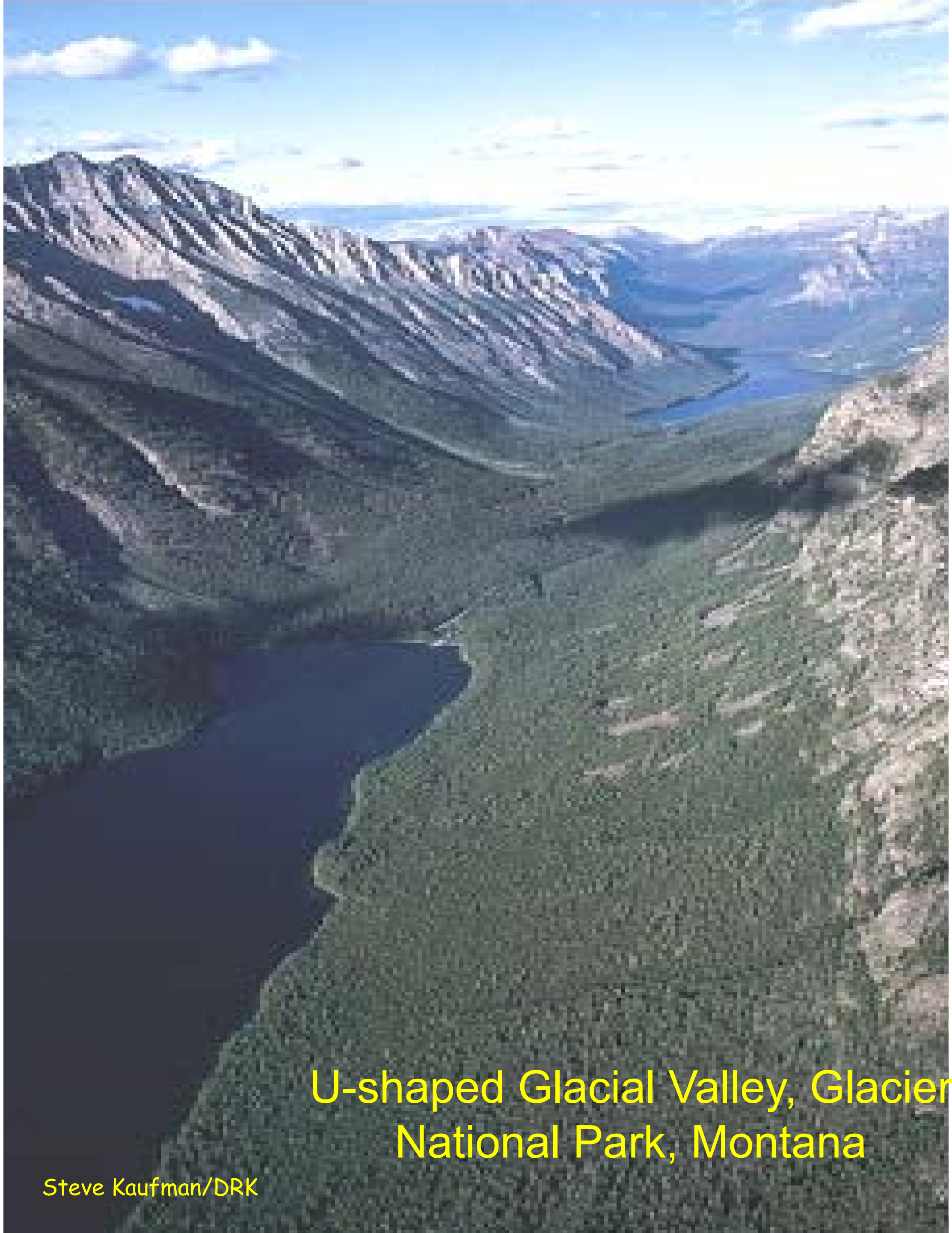
- Amphitheatre shaped
- Characterised by
  - Steep sided slope (3 sides)
  - Open end (one side)
  - Flat bottom

- May also develop into tarn lakes

Glacial Landforms

2. "U" shaped valley

- Uniform erosional activity
- Horizontal and vertical
- Steep sided and flat bottom

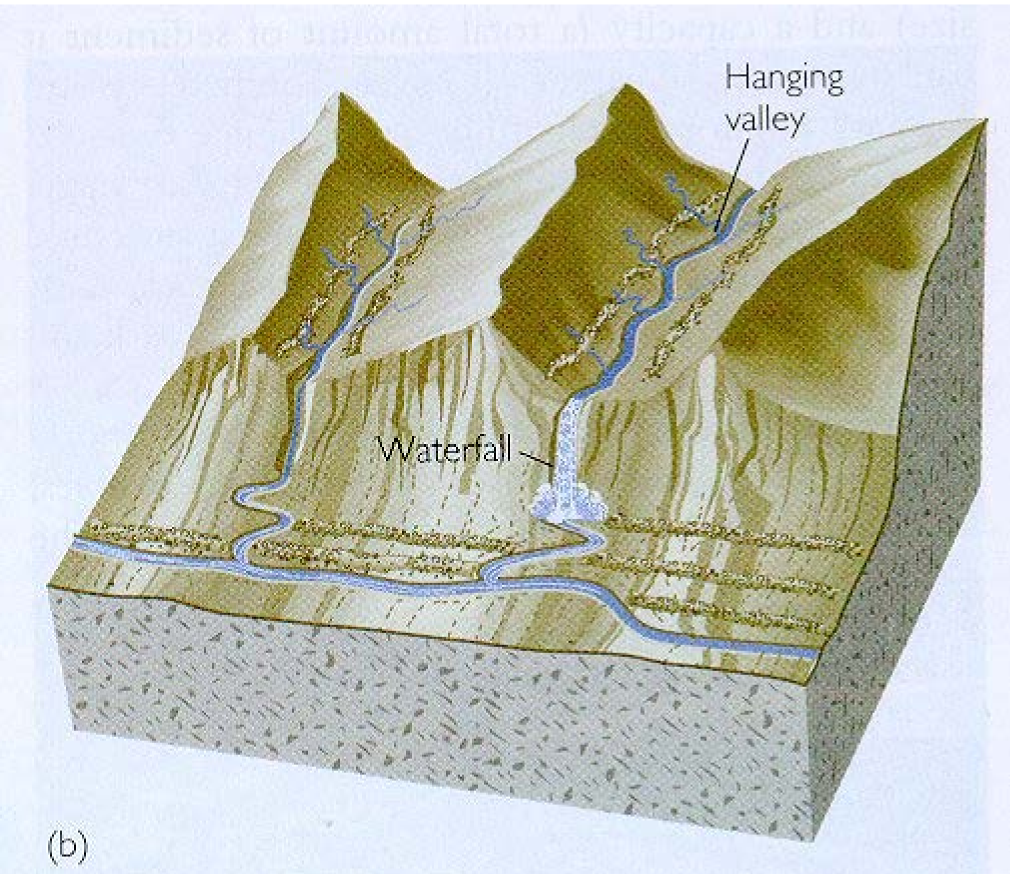


U-shaped Glacial Valley, Glacier National Park, Montana

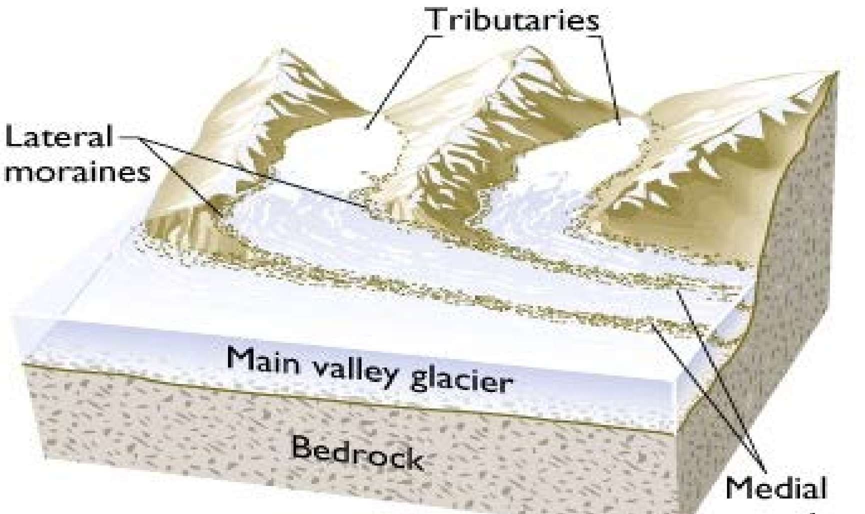
Fig. 15.20

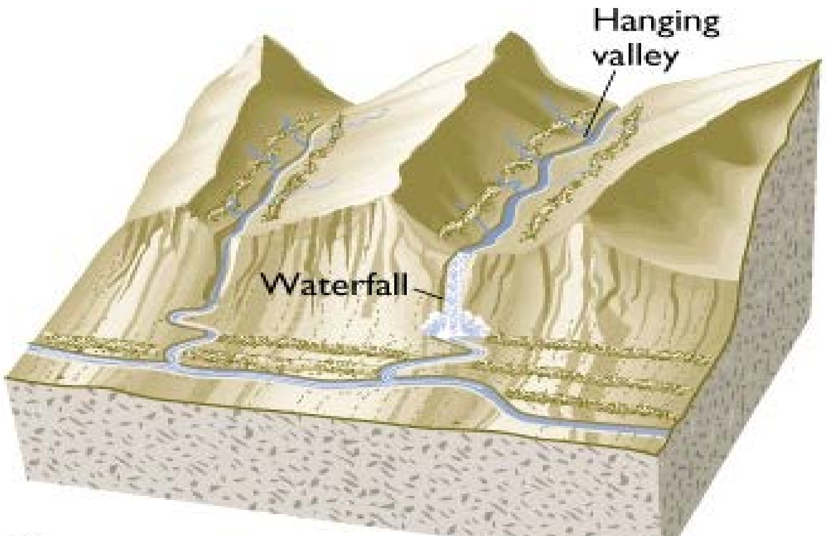
Glacial Landforms

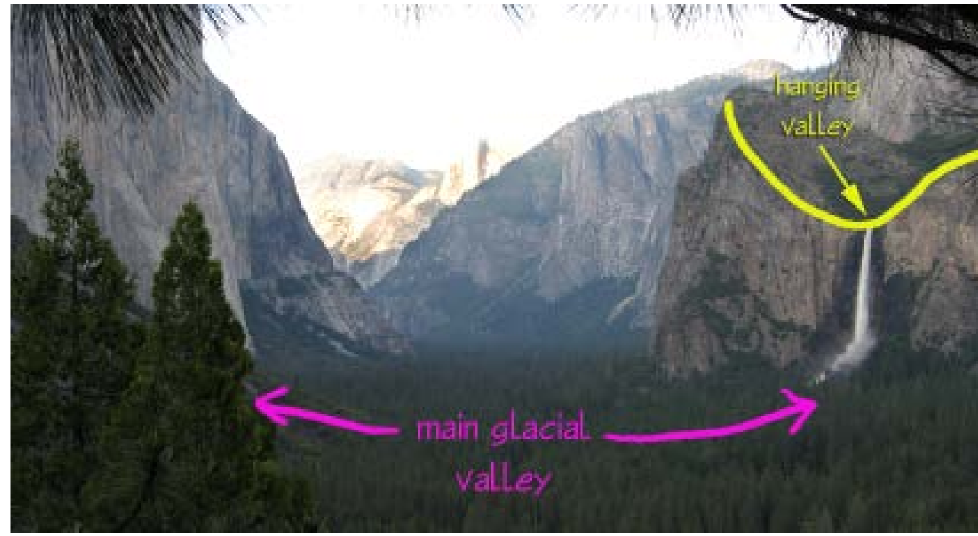
3. Hanging valley



- Smaller tributaries are unable to cut as deeply as bigger ones and remain “hanging” at higher levels than the main valley as discordant tributaries
- due to glacial tilting or faulting

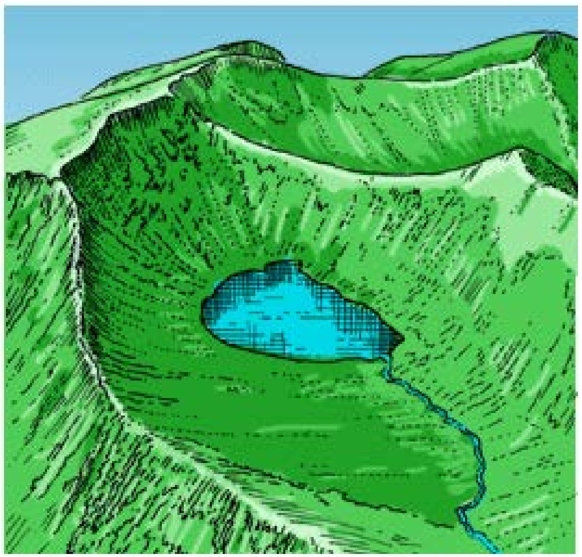

Tributaries
Lateral moraines
Main valley glacier
Bedrock
Medial moraines
(a)

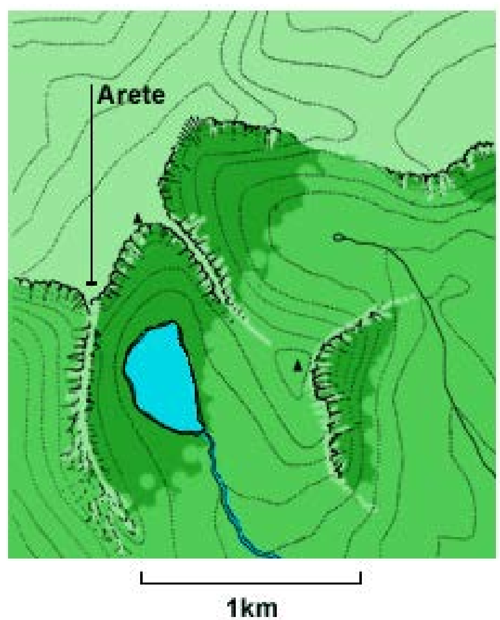

Hanging valley
Waterfall
(b)


hanging valley
main glacial valley

Glacial Landforms

4. Arete





- Steep sided
- Sharp tipped summit
- Typically formed when two glaciers erode parallel U-shaped valleys.

Glacial Landforms

5. Fjord

- A fjord is formed when a glacier retreats, after carving its typical U-shaped valley, and the sea fills the resulting valley floor.
- Narrow, steep sided inlet (sometimes deeper than 1300 metres) connected to the sea

6. Fjord

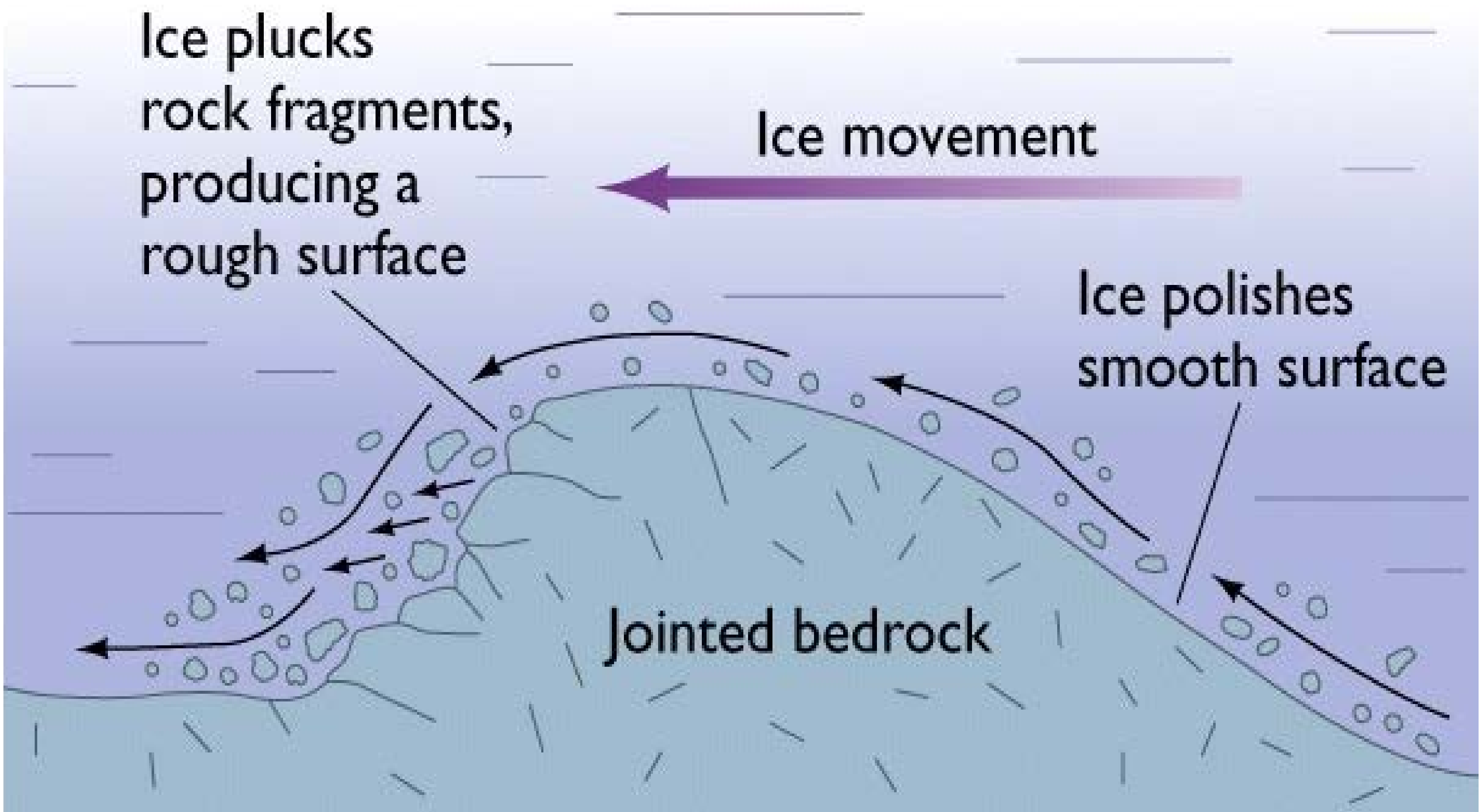
Glacial Landforms

6. Roche moutonnée

- If a glacier flows over a bed rock knob, it carves an elongate, streamlined hill.
- Assymetrical hill
- Characterised by
  - gently sloping: upstream, smooth, polished
  - Abrupt: downstream

Glacial Landforms

7. Roche moutonnée



Glacial Deposits

- Erratics
- Glacial drift
  - Till
  - Stratified drift

Glacial Deposits

- **Erratics**
  - Large boulder size rock type
  - Transported by glaciers

Glacial Deposits

Glacial drift

- Averages 6 m thick over the rocky hills
- Two types
  - Till:
    - deposited directly by glacial ice
    - Highly variable
    - Variable grain sizes
    - Sharp edged or irregular shaped pebbles and cobbles
  - Stratified drift:
    - carried by streams and deposited by streams
    - Distinct horizontal bands
    - Rounded

Glacial Deposits

Till

Stratified drift

Martin Miller

Landforms by Glacial Deposits

Moraine

- Accumulation of till deposited
- Based on the locations, characterised by several types.

  - End moraine
  - lateral moraine
  -medial moraine


Lateral
Moraines
Stephen J. Krasemann/DRK Photo

Landforms by Glacial Deposits

Drumlines

- An elongated hill formed when a glacier flows over and reshapes a mound of till or stratified drift.

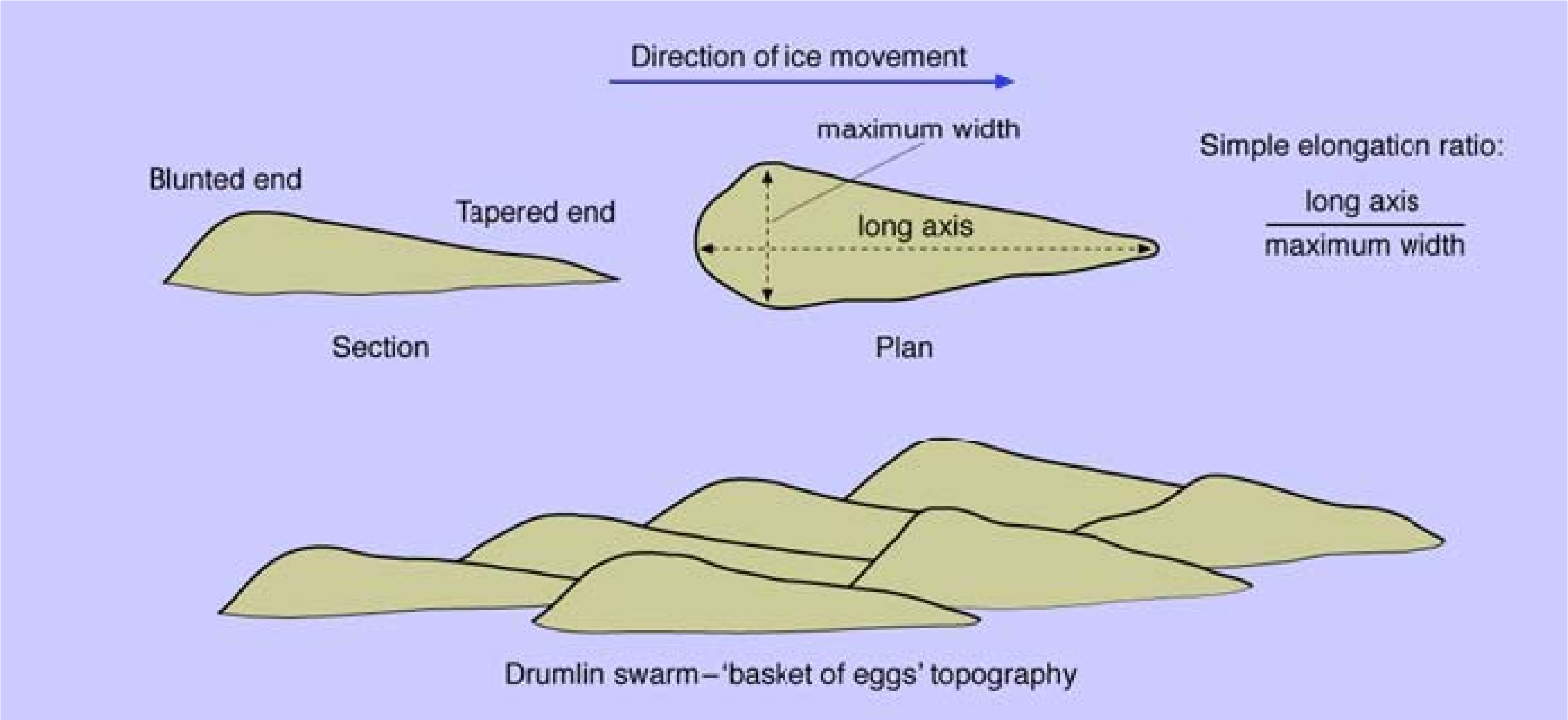
- Inverted spoon shape deposits.

- The erosional counterpart is called a roche moutonne.

Landforms by Glacial Deposits

Drumlines

- Trace past glacier movement



Tom Bean/DRK

Landforms by Glacial Deposits

Outwash

- When glacier reached its lowest point and melts, it leaves behind a stratified deposition material.

Landforms by Glacial Deposits

Esker

- A long narrow ridge, often sinous
- Stratified sediments
- Formed as a channel deposite of a stream that flowed within or beneth a melting glaciers


Tim Hauf Photography/Visuals Unlimited